Elbow junction

Course notes for first-year medical students.
Dr. Soussa Arbi C.
Laboratory of Medical and Surgical Anatomy
Faculty of Medicine - Annaba

I-Introduction

-Definition

The elbow is a complex joint that connects the humerus (upper arm bone) and the two forearm bones (radius and ulna). It consists of three diarthrose joints:

• Humero-ulnar joint: This is a trochlear joint in which the forearm flexes and extends on the upper arm.

• Proximal radio-ulnar joint: This is a trochoid joint, adapted for pronation and suspiration.

• Humero-radial joint: This is a ball-and-socket joint and participates in the movements of the other two.

The three joints are essentially one; in fact, they share a single joint cavity, a single synovial membrane, and a single ligamentous apparatus.

II-Articular surfaces

A-Articular surfac of the distal end of the humerus

It consists of :

1-Humeral trochlea ;this is a segment of a pulley,with two slopes and a groove .it is surmounted ;anteriorly by the coronoid fossa and poseriorly by the olercraneon fossa.

2-Capitumlum of the gumerus ;this is a segment of a spheroid .

Anterior and superior to the capitulum is a depression ,the radial fossa

3-Capitulum-trochlear groove :Located betwen the ccapitulum and the trochlea of the humerus (coronoid zone)

B- Articular surfaces of the upper extremity of the ulna

We distinguish two articular surfaces

-Trochlear notch of the ulna

- Radial incision

*** Trochlear notch of the ulna**

Has a hook shape that fits onto the surface of the humeral trochlea.

Presents two articular facets covered with cartilage

Separated from each other by a transverse groove.

The articular facets:

-Posterior, vertical, Formed by the anterior surface of the olecranon.

-Anterior, horizontal, Occupies the superior surface of the coronoid process

*** Radial incision**

Occupies the external face of the coronoid process,

In the shape of a hollow cylinder segment.

The radial notch articulates with the periphery of the head of the radius

C-Articular Surfaces of the Superior Extremity of the Radius

The head of the radius has Two articular surfaces:

- The articular fovea: the superior surface
- The articular circumference of the radius

1- Articular fovea

It is excavated and articulates with the capitulum of the humerus

Bounded by a rim whose inner part is beveled

Articulates with the capitulo-trochlear groove.

2- Articular circumference of the radius

Covered with articular cartilage, continuous superiorly with that of the articular fovea
Occupies the circumference of the radial head.
The circumference of the radial head is enclosed within an osteofibrous ring, formed by the radial notch and the annular ligament.

3-Annular ligament

It is a fibro-cartilaginous band 1 cm high which constitutes both an articular surface on its deep side and a passive ligament on its superficial side.
It extends from the anterior border to the posterior border of the radial notch of the ulna, wrapping around the radial head
The annular ligament has two faces **:**
-Inner surface (articular): related to the circumference of the radial head, covered with articular cartilage.
-Outer surface: fibrous, adhering to the articular capsule**.**

III- Means of union

1- Joint capsule

It is a fibrous sleeve that connects the three ends of the bones.
It attaches around the joint surfaces.
It is fixed:
- Superiorly on the humerus: it runs along the superior border of the coronoid and radial fossae anteriorly, and the olecranon fossa posteriorly.
- Laterally: on the inferior borders of the medial and lateral epicondyles.
- Inferiorly on the ulna: on the edges of the notches near the cartilage.
- Inferiorly on the neck of the radius: half a centimeter below the radial head.
- Its deep surface is lined by the synovial membrane.2-Ligaments

The joint capsule is reinforced by six ligaments:

- **Anterior ligament**

- Poorly defined
- Extends over the entire anterior portion of the joint capsule
- Among its bundles, the following are distinguished:
- The anterior oblique bundle, extending from the superior border of the coronoid fossa to the superior border of the coronoid process

- **Posterior ligament**

It consists of three bundles:
- **Oblique humero-olecranon bundles**: these extend from the lateral edges of the olecranon fossa to the apex of the olecranon.
- **Horizontal humero-humeral bundles:** these run from one edge of the olecranon fossa to the other, forming a transverse band.
- Vertical humero-olecranon bundles: these extend from the upper part of the olecranon fossa to the apex of the olecranon.

- **Ulnar collateral ligament**

It is formed of three bundles (anterior, middle, and posterior):
- **Anterior bundle:** extends from the medial epicondyle to the coronoid process.
- Middle bundle: thick and wide, it inserts superiorly on the medial epicondyle and inferiorly on the coronoid process.
- **Posterior bundle (Bardinet's bundle):** fan-shaped, attached at its apex to the medial epicondyle, and terminates on the olecranon.
- **Arcuate bundle (Cooper's bundle**): runs transversely from the coronoid process to the base of the olecranon

- **Radial collateral ligament**

Also composed of three bundles:

- Anterior bundle: extends from the lateral epicondyle to the radial notch**.**

- Middle bundle: inserts superiorly on the lateral epicondyle and inferiorly on the ulna (posterior to the radial notch).

- Posterior bundle: extends from the lateral epicondyle to the lateral border of the olecranon**.**

- **Inferior ligament (Quadratus denuccius)** It has the shape of a quadrilateral plate that extends from the inferior border of the radial notch to the medial part of the neck of the radius
- **Annular ligament**

IV- <u>Synovium</u>

-The synovium is a membrane that lines the inner surface of the joint capsule.
-It secretes synovial fluid.
-It reflects off the bone ends and stops where the articular cartilage begins.
-Thus forming recesses:

➢ An anterior recess: which corresponds to the coronoid and radial fossae.
➢ A posterior recess: related to the olecranon fossa.
➢ An inferior (annular) recess: surrounds the neck of the radius.

V- <u>Bursae</u>

- Bursae are periarticular structures whose function is to reduce friction between soft tissues and protruding bony structures.
Example: the olecranon bursa

VI- <u>Functional Anatomy</u>

The elbow joint can perform:
- Flexion and extension movements, which occur at the humero-antebrachial joint.
- Pronation and supination movements, which occur at the humeroradial and proximal and distal radioulnar joints.
- **<u>Flexion</u>**: Moves the forearm forward - Range of motion = 140° to 160°
- **<u>Extension</u>**: Moves the forearm backward - Range of motion = 0°
- Rotation of the forearm around a vertical axis (radius is mobile).
- **<u>Supination</u>**:Rotates the forearm outward, palm facing upward(Range of motion=85°).
- **<u>Pronation</u>**:Rotates the forearm inward,palm facing downward (Range of motion=70°).

<u>VII-Surface Anatomy (Palpatory Anatomy):</u>

The elbow has three palpable bony landmarks posteriorly:
1- Lateral: lateral epicondyle
2- Medial: olecranon
3- Medial: medial epicondyle
- These landmarks form a triangle in flexion, called Nélaton's triangle.
- And in extension, they form a straight line, Malgaigne's line.

<u>Conclusion</u>: The elbow is a complex joint that connects the humerus and the two forearm bones.
It consists of three articulations.
It is surrounded by bursae and reinforced by muscles